Team Name: Student Nexus

campusMate

One platform for campus life.

Marketability & Validation

DOCUMENT	VERSION	DATE	FOUNDED BY
Marketability & Validation	1.0	29 August 2026	Delton & Freddy

THE MARKET

Who this is for, and how many there are

For campusMate to work, the market behind it has to be big enough. We looked at the same students the product overview talks about. Not to add more features, but to check the numbers, and to see if this kind of trading is already happening at scale.

01 **A market that's big and in one place**

South Africa's public universities have over 1.2 million students across 26 institutions. UJ alone has more than 50,000 students on one campus. We don't need to build this market from scratch. It already exists, in one place, every day.

02 **A market that's already spending, just not with us**

Students already buy and sell through WhatsApp groups, Facebook Marketplace and noticeboards every week. campusMate isn't asking students to do something new. It's giving something they already do a safer, better home.

03 **A market that repeats every single year**

Final year students leave with things to sell and nowhere good to sell them. First years arrive needing everything and don't know where to look. This happens every year, on every campus, without fail.

04 **A market that can grow the same way everywhere**

What works at UJ isn't just a UJ thing. The same closed campus idea works at the other 25 public universities, and later at private colleges and TVET colleges too.

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VALIDATION

Why we think students will actually switch

> A market only becomes a business once real students pick it over what they already use for free.

campusMate isn't asking students to trust something brand new. It's asking them to trust a better version of something they already do. That's a smaller ask, and it's the one we're testing first, before we spend any money on growth.

Here are the four checks we're running, in order:

A **Talk to students before building for them**

We're running interviews and surveys with UJ students, buyers, sellers and tutors, before making product decisions. Not after.

B **Look at what's already happening**

We're checking existing WhatsApp groups and Facebook Marketplace pages to see how much is really being traded and at what prices. That way we're working from real numbers, not guesses.

C **Test the real sticking point**

We're running a small, verified group the same way the live product will verify people. This tells us if being closed and verified actually changes how people behave, or if it just sounds good on paper.

D **Keep using it separate from paying for it**

Anything that costs money, like a tutoring fee or a boosted listing, only gets tested once the free version is already working. That way the two answers never get mixed up.

THE HARDWARE

What it runs on, and what it can sense

campusMate isn't only an app. Part of what makes it useful is a small computer on campus that quietly does a few extra jobs in the background.

01 **Runs locally on a Raspberry Pi**

The whole platform can run on a single Raspberry Pi based on campus. It's cheap, it doesn't need a big server, and it can keep going even when the internet connection isn't great. That keeps the cost of running it low enough that one campus, or ten, doesn't need a big budget.

02 **Checks the weather and suggests what to sell**

The Pi keeps track of the local weather and temperature and uses that to recommend items that actually make sense right now. On a cold week it might push jackets and heaters. On a hot one it pushes fans and cold drinks. Sellers get seen at the right time, instead of just whenever they happened to list.

03 **Spots busy areas and points sellers there**

It can tell where a lot of students are gathered on campus at any given time. Instead of guessing where to go and sell, sellers get pointed toward the spots that actually have people in them.

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BOM SUMMARY

The campusMate first year implementation requires an estimated R7,992 in once off costs and R2,351 in monthly operating costs.

The largest physical hardware investment is the Raspberry Pi 5 (8GB) at R4,620, which provides the local computing infrastructure for the campusMate prototype. The platform's core digital infrastructure consists of transactional and email services.

Additional costs cover a part time student moderator, domain registration, CIPC company registration, name reservation and the initial campus launch and marketing campaign.

Based on the current estimates, the total projected operating cost for the first 12 months is approximately R36,204, calculated as:

R7,992 once-off costs + (R2,351 × 12 months) = R36,204

The SSL certificate carries no additional cost because it is provided through the selected hosting configuration.

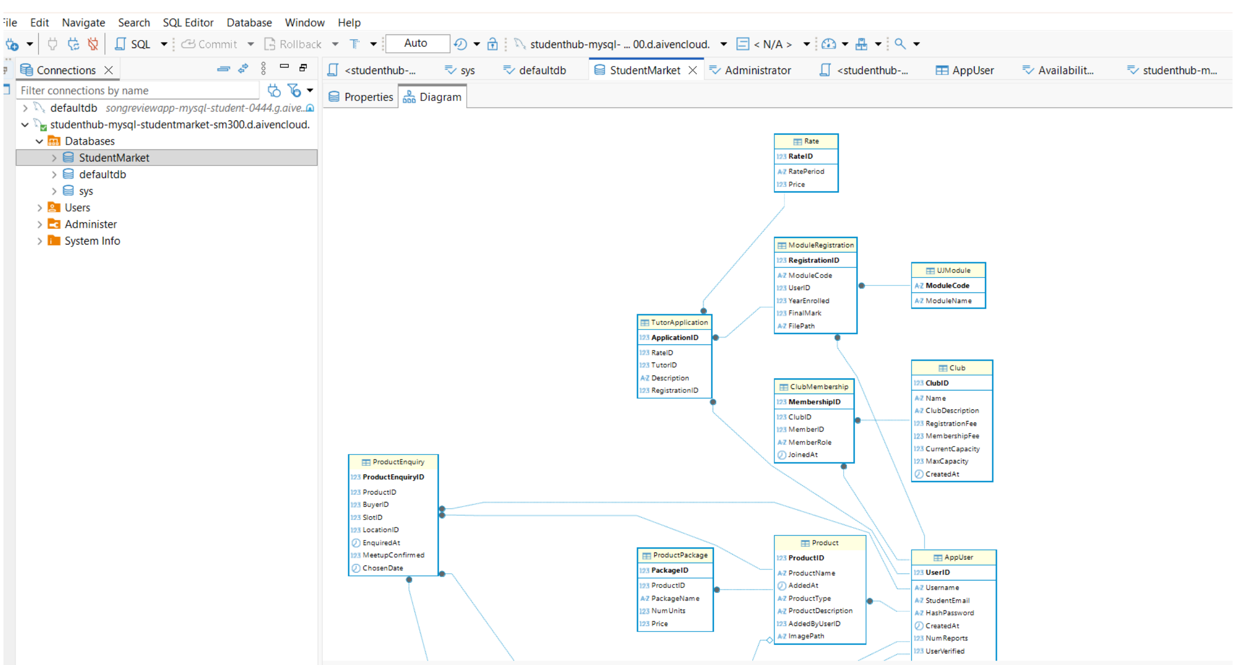
TECHNICAL IMPLEMENTATION

How campusMate is built and managed

campusMate isn't competing with another app. It's competing with habits that already work well enough to keep using, right up until something goes wrong.

CHANNEL	TECHNOLOGY USED	PLATFORM USED
Backend	c#, BCrypt	Visual Studio 2019
Frontend	HTML and CSS	Visual Studio 2019
Database	mySQL	Database currently hosted on Aiven, but we will be using raspberry pi once we get the components

Snippet of the database (schema)



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Snippet of the backend code (schema)

```
1 reference
public bool AccountExists(string usernameOrStudentEmail)
{
    bool isEmail = isStudentEmail(usernameOrStudentEmail);

    //sql query to find number of users with username or email
    string sql;
    if (isEmail)
    { //using student email
        sql = "SELECT COUNT(*) FROM AppUser WHERE StudentEmail = @Value";
    }
    else
    { //using username
        sql = "SELECT COUNT(*) FROM AppUser WHERE Username = @Value";
    }

    using var connection = new MySqlConnection(_connectionString);
    connection.Open();

    using var command = new MySqlCommand(sql, connection);
    command.Parameters.AddWithValue("@Value", usernameOrStudentEmail);
    long count = (long)command.ExecuteScalar();

    return count > 0;
}

//login method
1 reference
public User? Login(string usernameOrStudentEmail, string plainPassword)
{
    bool isEmail = isStudentEmail(usernameOrStudentEmail);
    string sql;
    if (isEmail) { //using student email
        sql = "SELECT UserID, Username, StudentEmail, HashPassword, CreatedAt, NumReports FROM AppUser WHERE StudentEmail = @Value";
    }
    else{ //using username
        sql = "SELECT UserID, Username, StudentEmail, HashPassword, CreatedAt, NumReports FROM AppUser WHERE Username = @Value";
    }

    using var connection = new MySqlConnection(_connectionString);
    connection.Open();

    using var command = new MySqlCommand(sql, connection);
    command.Parameters.AddWithValue("@Value", usernameOrStudentEmail);
    using var reader = command.ExecuteReader();

    if (!reader.Read())
    {
        return null; // no matching user, by email or username
    }

    string storedHash = reader.GetString("HashPassword");
```

Screenshot of the application's user profile